

Vrushabh Desai

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Experience

Cognex Corporation

14th Jun 2021 - Present

Software Engineer - Full-time

Natick, MA

- As a **Scrum Master**, guided team adherence to Agile principles, facilitating scrum events. Collaborated closely with the product owner, development team, and other stakeholders to ensure smooth project execution and delivery.
- Significantly contributed to creating a new vision library that combined various Cognex internal tools into one single API, which cut down deployment time for platform teams from one quarter to merely three weeks.
- Developed Python bindings for C++ vision library to enable rapid prototyping for end customers, accelerating development cycles for new advance tool.
- Led the documentation process for the new vision library, delivering detailed C++ and Python documentation with Python IntelliSense, resulting in an enhanced user experience.
- Enhanced the CI pipeline by integrating the Bullseye code coverage tool and developing a Blender-based app for generating synthetic barcode image, which improved code quality assessment and robust testing capabilities.

Earth Energy EV (Jindal Mobilirics)

11th Jun 2018 - 14th Jun 2019

Electronic Design Engineer - Full-time

Mumbai, India

- Led the design and development of a vehicle Electronic Control Unit (ECU), programming the embedded firmware in C. Integrated it with CAN protocol and peripherals to enhance the vehicle's communication systems.
- Managed the team of four members, responsible for creating a steering system for an autonomous delivery bot. Used Raspberry Pi 3, Open CV and machine learning algorithms for lane detection and obstacle avoidance.

DJS Racing (Formula Student Team)

20th Jun 2016 - 25th May 2018

Lead Electrical and Data Acquisition Engineer - Part-time

Mumbai, India

- Led team of seven, Played a key role in developing a power distribution module for an F-1 style race car.
- Designed a data logging and 3G Telemetry system using Race Capture pro MK3 to collect and store data from various sensors on the car. Developed a closed loop Drag Reduction System for Aerodynamic package.

Projects

Driver Activity Recognition | Python, Scikit Learn, OpenCV, TensorFlow

- Trained a model to identify distracted drivers and trigger an onboard alarm which helps alert the driver.
- Trained a multi-class classification algorithm and CNNs on 22.5k labelled images to identify human distractions into 10 different classes with 94% accuracy.

Detection and Recognition of Traffic Signals | C++, Python, OpenCV, TensorFlow

- Integrated YOLOv3 Deep Neural Network for robust object detection in the current frame
- Extracted image features using dynamic threshold and OpenCV libraries to detect traffic signals in a defined bounding box.

Technical Skills

Programming Language: Python, C, C++ (completed C++ Nano Degree), C#, Kotlin

Container Technology & Management, CI/CD Tools: Docker, Jenkins, JFrog, Conan

Operating Systems & Tools: Linux, Windows, ROS, Visual Studio, MATLAB, Atmel Studio

Education

MS in Robotics Engineering | Worcester Polytechnic Institute, Worcester, MA

May 2021

Relevant Course: Artificial Intelligence, Computer Vision, Motion Planning, Advance Robot Navigation.

CGPA: 4.00

BE in Electronics Engineering | D. J. Sanghvi College of Engineering, Mumbai, India

June 2018

Relevant Course: Embedded System Design, Control Systems, Operating Systems.

CGPA: 3.7

Achievements

- Awarded Best Designed Formula Student Car at Formula Bharat, India, consecutively in 2017 and 2018.
- Achieved Second Best Asian Team at Formula Student Germany 2017, Hockenheim Ring, Germany.
- Won Static Event (Cost) at Formula Student Austria held in July 2018 at Red Bull Ring, Austria.
- Fulfilled 200 hours of social service under the National Service Scheme during my undergraduate studies.